

TRILOKESH RANJAN SARKAR

AI/ML Engineer | Agentic AI & LLM Systems | Generative AI | NLP | Data Engineering

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SUMMARY

AI/ML Engineer with 2 years of experience building LLM-powered automation systems, OCR/NLP pipelines, and computer vision solutions across Retail, Healthcare, and BFSI. Experienced designing production workflows that combine LLM reasoning, tool-calling to external data sources, human-in-the-loop validation, and automated error-correction — reducing manual effort by up to 80% and improving operational scalability. Strong foundation in adversarial machine learning and model robustness, evidenced by a peer-reviewed arXiv publication. Comfortable across the AI stack: Python, SQL, PyTorch-based CV models, Generative AI (GPT-4o), NLP, and cloud data infrastructure (AWS, Redshift). Seeking to apply this experience to AI/ML Engineering or Data Scientist roles building agentic and production LLM systems.

TECHNICAL SKILLS

Languages & Core: Python, SQL, Regex

Generative AI / LLM: GPT-4o, Prompt Engineering, LLM-driven workflow design, Tool-calling, Human-in-the-loop (HITL) validation, Fine-tuning (resource-constrained), LLM output evaluation

Machine Learning / CV: YOLOv8, Computer Vision, Deep Learning, Image Classification, Transfer Learning (pre-trained models), PyTesseract OCR

Adversarial ML & Model Security: Adversarial Robustness Testing, FGSM, Carlini-Wagner (CW) Attacks, Defensive Distillation, Teacher-Student Model Distillation, CNN Architectures, Model Evaluation & Benchmarking

NLP: Named Entity Recognition (NER), Intelligent Text Parsing, OCR/NLP Automation, Regex-based Extraction

LLM / GenAI Engineering: GPT-4o, LLM API Integration, Prompt Engineering, LLM-driven Workflow Design, Tool-calling, Human-in-the-loop (HITL) Validation, Resource-constrained Fine-tuning, LLM Output Evaluation

Data Engineering & Cloud: AWS, Redshift, PostgreSQL, Apache Airflow (workflow orchestration), Label Studio (data annotation), ETL Pipelines

Visualization / BI: Power BI, Looker Studio, Interactive Dashboarding, Business Intelligence, KPI Reporting

Datasets & Benchmarks: Tiny-ImageNet, CIFAR-10

EXPERIENCE

Data Analyst

10/2024 – Present

Web Spiders | Kolkata, West Bengal

- LLM-powered automation pipeline:** Designed and orchestrated an LLM-driven validation system (GPT-4o) with tool-calling to external data sources, automated web research, and human-in-the-loop review checkpoints — built on Apache Airflow, AWS, Redshift, PostgreSQL, and Label Studio — to automate retail taxonomy validation and ontology auditing, improving data quality and governance at scale.
- Structured extraction with validation routing:** Built OCR/NLP-based extraction pipelines with confidence-based routing to human review, using Named Entity Recognition (NER) and intelligent text parsing to digitize handwritten and scanned documents — improving data accuracy and reducing manual effort by 80%.
- Handwriting OCR pipeline:** Built a custom PyTesseract-based handwriting OCR pipeline to digitize handwritten medical documents, layering NLP-based NER and regex matching to cut human intervention in data extraction by 80%.
- Computer vision — ANPR system:** Developed an Automatic Number Plate Recognition (ANPR) system using YOLOv8 and deep learning-based computer vision for real-time, high-accuracy license plate detection and recognition.
- Computer vision — in-store analytics:** Deployed pre-trained models for sentiment analysis of in-store customer facial expressions and built heat maps from video analytics to identify high-traffic areas and peak hours, informing store layout and staffing decisions.
- Analytics & BI:** Used SQL, Python, and Power BI to analyze trends, build interactive dashboards, and generate actionable insights supporting strategic planning and decision-making; built marketing analytics dashboards tracking email campaign KPIs that improved engagement 15% and conversions 12%.

Tech Stack: Python, SQL, GPT-4o, Power BI, AWS, Apache Airflow, YOLOv8, Regex, NLP

- **Adversarial robustness research:** Evaluated the security of a Generative AI model against adversarial threats by comparing FGSM and Carlini-Wagner attacks, and assessed distillation as a defense mechanism.
- **Model fine-tuning under constraints:** Improved model robustness and efficiency through optimized LLM fine-tuning under limited compute resources.

PUBLICATION

Comparing FGSM, Carlini-Wagner Attacks, and the Function of Distillation as a Defense Mechanism to Assess Adversarial Robustness

arXiv.org — 2023–2024 | <https://arxiv.org/abs/2404.04245>

- **Scope:** Technical study of adversarial attacks on deep neural networks used for image classification, evaluating FGSM and Carlini-Wagner (CW) attacks against three pre-trained classifiers — ResNeXt-50, DenseNet-201, and VGG-19 — on the Tiny-ImageNet dataset.
- **Defense mechanism:** Proposed and evaluated defensive distillation as a countermeasure on CIFAR-10, using ResNet-101 as the teacher model and ResNeXt-50 as the student model.
- **Findings:** Showed the distillation-based defense was effective against FGSM but remained partially vulnerable to the more sophisticated CW attack, with results validated through rigorous experimentation.
- **Recognition:** Peer-reviewed after review by GenAI researchers; contributes an evaluation framework comparing model vulnerabilities and defense effectiveness, offering actionable insights for secure model deployment.

EDUCATION

Post-Graduation Program in Data Science — Praxis Business School

2023 – 2024 | Kolkata, West Bengal | CGPI: 6.2 / 8

Bachelor of Computer Application — B. P. Poddar Institute of Management and Technology

2020 – 2023 | Kolkata, West Bengal | CGPA: 8.8 / 10

CERTIFICATIONS

- Databricks Fundamentals Accreditation — Databricks
- Generative AI Fundamentals Accreditation — Databricks
- SQL Certificate — HackerRank